APTRANSCO

Assistant Executive Engineer (Electrical) 2026

Master Formula & Revision Handbook

CBT Examination | 100 Marks | No Negative Marking

Exam Date: 26 August 2026

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High-Density • Exam-Focused • Last-Minute Revision Ready

OFFICIAL EXAM PATTERN & STRUCTURE

SECTION	TOPICS	MARKS	STATUS
Core Electrical Engineering	70	70	Primary
1. Electric Circuits	DC/AC/Networks/3-Phase/Harmonics	10	
2. Electrical Machines	Transformer/DC/Induction/Sync	15	
3. Power Systems	Generation/Transmission/Protection/Faults	18	
4. Utilization & Control	Heating/Traction/Control Systems	12	
5. Measurements	Bridges/Instruments/Measurement	8	
6. Analog & Digital Electronics	Diodes/BJT/FET/Logic	8	
7. Power Electronics & Drives	Rectifiers/Choppers/Inverters/SCR	9	
Common Section	30	30	Secondary
General Intelligence & Reasoning	5	5	
General Awareness	7	7	
English Language & Comprehension	5	5	
Quantitative Aptitude	8	8	
Computer Knowledge	5	5	
TOTAL	100	100	No negative marking

✓ Strategy: No negative marking = Attempt ALL questions | Accuracy > Speed

MASTER CONSTANTS & UNIT CONVERSIONS

CONSTANT	SYMBOL	VALUE	UNIT
π	π	3.14159 / 22/7	dimensionless
√2	√2	1.414	dimensionless
√3	√3	1.732	dimensionless
1/√3	1/√3	0.577	dimensionless
Permeability of free space	μ₀	4π × 10<super>-7</super>	H/m
Permittivity of free space	ε₀	8.854 × 10<super>-12</super>	F/m
Speed of light	c	3 × 10<super>8</super>	m/s
Charge of electron	e	1.602 × 10<super>-19</super>	C

SI PREFIX CONVERSIONS

tera (T)	10<super>12</super>	milli (m)	10<super>-3</super>
giga (G)	10<super>9</super>	micro (μ)	10<super>-6</super>
mega (M)	10<super>6</super>	nano (n)	10<super>-9</super>
kilo (k)	10<super>3</super>	pico (p)	10<super>-12</super>

ANGLE CONVERSIONS

1 radian = 57.3° | 1° = 0.01745 rad | 180° = π rad

1. ELECTRIC CIRCUITS - FORMULAS & RELATIONSHIPS

DC CIRCUITS FUNDAMENTALS

Ohm's Law ■■■

Formula: $V = IR$

Where: V=Voltage (V), I=Current (A), R=Resistance (Ω)

Use when: Always

Important: Ohm's law holds only for ohmic materials

■■ **Trap:** None

★ **Shortcut:** Linear resistances only

Power (DC) ■■■

Formula: $P = VI = I^2R = V^2/R$

Where: P=Power (W)

Use when: All DC circuits

Important: Choose formula based on given data

■■ **Trap:** ■■ Don't confuse $P = VI$ with $P = I^2R$

★ **Shortcut:** Use $P = I^2R$ when only I and R known

Energy ■■■

Formula: $E = Pt = VIt$

Where: E=Energy (J), t=time (s)

Use when: Energy dissipation

Important: P must be constant

■■ **Trap:** None

★ **Shortcut:** W = Joules

Resistance - Series ■■■

Formula: $R_{eq} = R_1 + R_2 + R_3...$

Where: All in Ω

Use when: Series circuits

Important: Voltages add, current same

■■ **Trap:** None

★ **Shortcut:** Highest resistance dominates

NETWORK THEOREMS

Thevenin's Theorem ■■■

1. Remove load, find open-circuit voltage = V_{th}
2. Short all sources, find equivalent resistance = R_{th}
3. Load sees: V_{th} in series with R_{th}

$I_{load} = V_{th}/(R_{th} + R_{load})$

■■ **Trap:** Forgetting to remove load when calculating V_{th}

Norton's Theorem ■■■

1. Short load, find short-circuit current = I_n
2. Find equivalent resistance = R_n (same as R_{th})
3. Load sees: I_n in parallel with R_n

$I_{load} = I_n \times R_n/(R_n + R_{load})$

Maximum Power Transfer (DC) ■■■

Condition: $R_{load} = R_{Th}$

Maximum Power: $P_{max} = V_{Th}^2/(4 \times R_{Th})$

■■ **Trap:** In AC, matching is complex conjugate, not just equal

★ **Shortcut:** When $R_{load} = R_{Th}$, efficiency = 50%

AC FUNDAMENTALS & RMS VALUES

PARAMETER	SINUSOIDAL WAVE	FORMULA	MEANING
Peak Value	A_m	Maximum amplitude	Highest instantaneous value
RMS Value	V_{rms}	A_m/√2 = 0.707 × A_m	Equivalent DC heating value
Average Value	V_{avg}	(2/π) × A_m = 0.637 × A_m	Mean of half-cycle
Form Factor	FF	V_{rms}/V_{avg} = 1.11	For sinusoid always 1.11
Peak Factor (Crest)	CF	A_m/V_{rms} = √2 ≈ 1.414	Ratio of peak to RMS
Phase angle	ϕ	tan<super>-1</super>(X/R)	Lead (capacitive) or Lag (inductive)
Impedance	Z	√(R<super>2</super> + X<super>2</super>)	Total opposition in Ω
Reactance (Inductive)	X_L	2πfL = ωL	Inductive opposition in Ω
Reactance (Capacitive)	X_C	1/(2πfC) = 1/(ωC)	Capacitive opposition in Ω

■■ **Common Trap:** RMS vs Average values - RMS = 1.11 × Average for sine wave

★ **Quick Memorize:** For sinusoid: RMS = 0.707 × Peak, Avg = 0.637 × Peak

RLC CIRCUITS - RESONANCE & Q-FACTOR

Series RLC Resonance ■■■

Resonant Frequency: $f_r = 1/(2\pi\sqrt{LC})$ Hz

Angular frequency: $\omega_r = 1/\sqrt{LC}$ rad/s

At resonance: $X_L = X_C$, Impedance = R (minimum)

Current at resonance: $I_{max} = V/R$ (maximum)

Quality Factor: $Q = \omega L/R = 1/(\omega RC) = f_r/BW$

Bandwidth: $BW = f_r/Q = R/(2\pi L)$

■■ **Trap:** Series resonance has MIN impedance; Parallel resonance has MAX impedance

Parallel RLC Resonance ■■

Same resonant frequency: $f_r = 1/(2\pi\sqrt{LC})$

At resonance: Total current is minimum, Impedance is maximum

For parallel: $Q = R \times \omega C = R/(\omega L) = \omega RC$

★ **Shortcut:** Parallel impedance at resonance ≈ L/RC (for high Q)

THREE-PHASE CIRCUITS

Star (Y) Connection ■■■

Line Voltage: $V_L = \sqrt{3} \times V_{phase}$
Line Current: $I_L = I_{phase}$ (same)
Total Power: $P = \sqrt{3} \times V_L \times I_L \times \cos\phi$ (W)
Reactive Power: $Q = \sqrt{3} \times V_L \times I_L \times \sin\phi$ (VAR)
Apparent Power: $S = \sqrt{3} \times V_L \times I_L$ (VA)

Delta (Δ) Connection ■■■

Line Voltage: $V_L = V_{phase}$ (same)
Line Current: $I_L = \sqrt{3} \times I_{phase}$
Total Power: $P = \sqrt{3} \times V_L \times I_L \times \cos\phi$ (W) [Same as Star]

Star to Delta Conversion

$Z_\Delta = 3 \times Z_Y$ (impedance)
$I_\Delta = I_Y/\sqrt{3}$ (current)

■■ **Trap:** Line quantities are different in Star vs Delta for impedance

★ **Remember:** 3-phase power formula is SAME for both: $P = \sqrt{3}\, VI\cos\phi$

2. ELECTRICAL MACHINES - TRANSFORMERS & MOTORS

TRANSFORMERS

EMF Equation (Faraday's Law) ■■■

$E = 4.44 \times f \times N \times \Phi_m$ (Volts)

Where: f = frequency (Hz), N = turns, Φ_m = max flux (Wb)

Transformation Ratio: $K = V_s/V_p = N_s/N_p = I_p/I_s$

For ideal transformer: $V_p/V_s = N_p/N_s = I_s/I_p$

Voltage Regulation (Approx.) ■■■

$VR\% \approx (I_2R_2\cos\phi \pm I_2X_2\sin\phi) / V_2 \times 100$

+ for lagging load (inductive)

– for leading load (capacitive)

■■ **Trap:** Sign reversal for leading vs lagging PF causes most mistakes!

★ **Shortcut:** For minimum regulation, make ($X\cos\phi l = R\sin\phi l$) at that load angle

Transformer Efficiency ■■

$\eta\% = (\text{Output Power} / \text{Input Power}) \times 100$

$\eta\% = (P_{out} / P_{out} + \text{Copper Loss} + \text{Core Loss}) \times 100$

Maximum Efficiency occurs when: Copper Loss = Core Loss

At max efficiency: $\eta_{max} = \sqrt{(1 - P_{core}/S_{rated})}$

Open Circuit (OC) & Short Circuit (SC) Tests

OC Test (low voltage side open):

- Core loss $= W_{oc} = V_{oc} \times I_{oc} \times \cos\phi_{oc}$
- Magnetizing reactance $X_m = V_{oc} / I_m$

SC Test (low voltage side shorted):

- Copper loss $= W_{sc} = I_{sc}^2 \times R_{eq}$
- Equivalent impedance: $Z_{eq} = V_{sc}/I_{sc}$

DC MACHINES (GENERATORS & MOTORS)

EMF Generated (Generator) ■■■

$E = PN\Phi/60A$ (Volts)

Where: P = poles, N = speed (rpm), Φ = flux/pole (Wb), A = parallel paths

Simplified: $E = K_e \times N \times \Phi$, where $K_e = PZ/(60A)$

Torque Produced (Motor) ■■■

$T = P\Phi I_a Z/(60A)$ (N-m)

Simplified: $T = K_t \times \Phi \times I_a$, where $K_t = PZ/(60A)$

■■ **Trap:** Torque proportional to flux × armature current, NOT speed

Back EMF (Motor) ■■

$E_b = V_t - I_a R_a$ (Volts)

Where: V_t = terminal voltage, I_a = armature current, R_a = armature resistance

Back EMF ∝ Speed: $E_b = K_e \times N \times \Phi$

Speed Equation (Motor) ■■■

$N = (V_t - I_a R_a) / (K_e \times \Phi)$

For shunt motor: $N \propto (V_t - I_a R_a)/\Phi$

★ **Shunt Motor characteristics:** Constant speed (flux constant), varying torque

★ **Series Motor characteristics:** High starting torque, high starting current, variable speed

THREE-PHASE INDUCTION MOTOR

Synchronous Speed ■■■

N_s = 120f/P (rpm)

Where: f = frequency (Hz), P = number of poles

At 50 Hz: 2-pole = 3000 rpm, 4-pole = 1500 rpm, 6-pole = 1000 rpm, 8-pole = 750 rpm

Slip ■■■

s = (N_s - N_r)/N_s = (N_s - N_r)/N_s

Rotor Speed: N_r = N_s(1 - s)

Rotor Frequency: f_r = s × f

■■ **Trap:** Slip is fraction (0.05 = 5%), NOT percentage when used in formulas

★ **At no-load:** s ≈ 0.5-2%, At full-load: s ≈ 3-5%

Torque (Rotor) ■■■

$T = 3 \times V^2 \times R_r / (2\pi N_s \times (R_r^2 + X_r^2/s^2))$

Condition for Maximum Torque: $s = R_r/X_r$

Maximum Torque: $T_{max} = 3V^2 / (4\pi N_s X_r)$

★ **Starting torque:** $T_{start} = T_{max} \times 2sR_r/(R_r^2 + X_r^2)$

Power Flow

Air-gap Power: $P_{ag} = (3 \times I_r^2 \times R_r) / s$

Rotor Copper Loss: $P_{cu} = s \times P_{ag}$

Mechanical Power: $P_{mech} = (1-s) \times P_{ag}$

■■ **Trap:** As slip increases, copper loss increases, mechanical power decreases

3. POWER SYSTEMS - TRANSMISSION, FAULTS & PROTECTION

GENERATION - LOAD FACTORS & CAPACITY

FACTOR	FORMULA	RANGE	MEANING
Load Factor (LF)	Avg Load / Peak Load	0.3-0.8	Utilization of capacity
Demand Factor (DF)	Max Simultaneous Demand / Connected Load	<1	Max simultaneous use
Diversity Factor (Div F)	Sum of Individual Max / Overall Max	>1	Non-coincident peaks
Capacity Factor (CF)	Avg Output / Rated Capacity	0.3-0.5	Plant utilization over time
Plant Factor (PF)	Actual Energy / Possible if 100% utilization	<1	Overall effectiveness

TRANSMISSION LINES - VOLTAGE REGULATION & EFFICIENCY

Short Line (< 80 km) ■■

$V_s = V_r + I_r(R \cos\phi + X \sin\phi)$

Regulation: $VR = I_r(R \cos\phi + X \sin\phi) / V_r$

Efficiency: $\eta = P_r / (P_r + I_r^2 R) \times 100$

Medium Line (80-250 km) - Nominal T Model ■

Equivalent Series impedance at mid-point

Receiving end voltage accounts for half capacitance effect

Long Line (> 250 km) - Distributed Parameters ■

Use ABCD parameters for sending/receiving-end relationships

Surge Impedance: $Z_c = \sqrt{(L/C)}$

Surge Impedance Loading (SIL) $= V^2/Z_c$

Ferranti Effect: $V_r > V_s$ at no-load (due to line capacitance)

Corona & Voltage Gradient ■

Critical Disruptive Voltage: $V_{cd} = 30 \times \delta \times r \times \ln(d/r)$ kV

Where: δ = air density factor, r = conductor radius (cm), d = separation (cm)

Corona Loss: $P = 241 \times 10^{-5} \times (f + 25) \times r \times (V - V_{cd})^2/\log(d/r)$ kW/km

■■ **Trap:** Corona loss increases exponentially with voltage above critical

PER-UNIT (PU) SYSTEM

Base Values Selection:

- Choose arbitrary S_{base} (usually system rating)

- Choose V_{base} for primary voltage

- Base current: $I_{base} = S_{base} / (\sqrt{3} \times V_{base})$
- Base impedance: $Z_{base} = V_{base}^2 / S_{base} = V_{base} / I_{base}$

For transformers (secondary side base values):

- S_{base} (unchanged)

- $V_{base2} = V_{base1} \times (T_2/T_1)$
- $Z_{base2} = V_{base2}^2 / S_{base}$

Per-unit value: Quantity (pu) = Actual value / Base value

★ **Advantage:** Transformer impedance same on both sides in PU

POWER FACTOR CORRECTION

kVAR Required for PF Correction ■■■

$Q_c = P(\tan\phi_1 - \tan\phi_2)$ kVAR

Where: ϕ_1 = initial angle, ϕ_2 = final angle

Capacitor Rating: $C = Q_c / (2\pi f V^2)$ Farads

Direct Method (more common):

If P = 100 kW, $\cos\phi_1 = 0.6$ (lagging), target $\cos\phi_2 = 0.9$ (lagging):

Find $\tan(\cos^{-1}0.6) \approx 1.33$ and $\tan(\cos^{-1}0.9) \approx 0.484$

$Q_c = 100(1.33 - 0.484) = 84.6$ kVAR

★ **Shortcut table:** Common $\tan\phi$ values - cos60° = 0.5, tan=1.73; cos45° = 0.707, tan=1

SYMMETRICAL COMPONENTS & FAULT ANALYSIS

Decomposition of 3-Phase to Symmetrical Components

$V_a = V_0 + V_1 + V_2$
$V_b = V_0 + \alpha^2 V_1 + \alpha V_2$
$V_c = V_0 + \alpha V_1 + \alpha^2 V_2$
Where: $\alpha = e^{j2\pi/3} = -0.5 + j0.866$, $\alpha^2 = -0.5 - j0.866$, $\alpha^3 = 1$

Three-Phase (Balanced) Fault ■■

$I_f = V_f / Z_1$ (uses positive sequence only)
Fault MVA = $(V^2 \times MVA_{base}) / Z_1$ (pu base)

Single-Line-to-Ground (LG) Fault ■

$I_f = 3V_f / (Z_0 + Z_1 + Z_2)$
Usually: $I_{LG} > I_{3\text{-phase}}$ (can be 1.5-2 times)

Line-to-Line (LL) Fault ■

$I_f = \sqrt{3} \times V_f / (Z_1 + Z_2)$
Magnitude: $I_{LL} \approx 0.87 \times I_{3\text{-phase}}$

Double Line-to-Ground (LLG) Fault ■

$I_f = 3V_f / (Z_1 + (Z_0||Z_2))$
■■ **Trap:** Fault current depends on sequence impedances, NOT just positive

★ **Memorize:** LG > LLG ≈ 3-phase > LL (in most systems)

PROTECTION RELAYS & CIRCUIT BREAKERS

Overcurrent Relay (IDMT - Inverse Definite Minimum Time) ■■

Operating Time: $t = (0.14 / (M^n - 1)) \times T_d$ [IEC standard]

Where: $M = I/I_{set}$ (fault current ratio), T_d = dial setting, n = characteristic

Plug Setting: $I_{set} = PS \times I_{CTprimary}$ / CT ratio

Example: PS = 0.8, CT = 400/5 ■ $I_{set} = 0.8 \times 400 = 320$ A

Percentage Differential Relay ■

Differential Current: $I_d = |I_{primary} - I_{secondary}|$

Bias Current: $I_b = (|I_p| + |I_s|) / 2$

Operating condition: $I_d > k \times I_b + I_{min}$

Where: k = 0.3-0.5 (slope %), I_{min} = 0.2-0.5 A

★ **Protection:** Operates on current difference (fault detection inside zone)

Distance Relay (Impedance Type) ■

Operating condition: $Z = V/I < Z_{set}$

Mho Relay (circular characteristic): $|V - Z_{set} \times I| < Z_{set} \times I$

Reactance Relay: $Im(V/I) < X_{set}$

■■ **Trap:** Mho relay most affected by power swings, need stability check

Circuit Breaker Recovery & Restriking Voltage

RRRV = Rate of Rise of Recovery Voltage (V/µs)

Determines CB's ability to interrupt high fault currents

CB must interrupt before peak arc voltage exceeds recovery voltage

4. UTILIZATION & CONTROL SYSTEMS

CONTROL SYSTEMS - TRANSFER FUNCTIONS & STABILITY

Standard Transfer Functions & Time Response ■■■

SYSTEM TYPE	TRANSFER FUNCTION	STEP RESPONSE	STEADY-STATE ERROR
Type 0 (Proportional)	1/(1+Ts)	Exponential rise	K/(1+K_p)
Type 1 (Integral)	1/[s(1+Ts)]	Ramp	Zero
Type 2	1/[s<super>2</super>(1+Ts)]	Parabola	Zero
2nd Order (Underdamped)	ω_n<super>2</super>/(s<super>2</super>+2ζω_ns+ω_n<super>2</super>)	Oscillatory	Depends on input

Steady-State Error Calculation ■■

$e_{ss} = \lim[s\to 0]\ s \times E(s) = \lim[s\to 0]\ s \times R(s)/(1+G(s)H(s))$

For Step Input ($R(s) = 1/s$): $e_{ss} = 1/(1+K_p)$, $K_p = G(0)$

For Ramp Input ($R(s) = 1/s^2$): $e_{ss} = 1/K_v$, $K_v = \lim[s\to 0]\ sG(s)$

For Parabolic Input ($R(s) = 1/s^3$): $e_{ss} = 1/K_a$, $K_a = \lim[s\to 0]\ s^2G(s)$

Routh-Hurwitz Stability Criterion ■■

For polynomial: $a_0s^n + a_1s^{n-1} + \ldots + a_n = 0$

Condition: All coefficients positive AND all 1st column entries positive in Routh array

Number of roots in right half-plane = Number of sign changes in 1st column

★ **Quick check:** If any coefficient missing or negative ■ unstable

Bode Plot - Gain & Phase Margins ■■

Gain Margin: $GM = -20\log|G(j\omega_{pc})|$ dB at phase crossover ($\phi = -180°$)

Phase Margin: $PM = 180° + \angle G(j\omega_{gc})$ at gain crossover ($|G| = 0$ dB)

Stability: Both GM > 0 AND PM > 0

■■ **Trap:** GM and PM reversed definitions cause many errors

★ **Typical requirement:** GM ≥ 6 dB, PM ≥ 30-45°

Root Locus - Key Concepts ■■

Properties:

- # of branches = # of poles
- # of asymptotes = # of poles - # of zeros
- Asymptote angles: $(2k+1)180°/(n-m)$, $k=0,1,...,n-m-1$
- Asymptote centroid: $\Sigma(\text{poles}) - \Sigma(\text{zeros}) / (n-m)$
- Breakaway points: $dK/ds = 0$ where $K = -1/G(s)H(s)$
- Angle condition: $\angle G(s)H(s) = (2k+1)180°$ on root locus

Compensators - Lag, Lead, Lead-Lag ■

Lag Compensator: $G_c(s) = (1+T_1 s)/(1+\alpha T_1 s)$, $\alpha > 1$

- Improves steady-state accuracy
- Reduces bandwidth & response speed

Lead Compensator: $G_c(s) = (1+T_2 s)/(1+\beta T_2 s)$, $\beta < 1$

- Improves transient response & stability
- Increases bandwidth & noise sensitivity

ELECTRIC TRACTION (TRAINS & TROLLEYS)

Speed-Time Curve Analysis ■■

Curve has three stages:

1. **Acceleration phase:** Constant tractive force F, varying speed

Acceleration: $a = F/m$

Energy: $E_1 = ½m(V_{max})^2$ (kinetic energy)

2. **Coasting phase (optional):** No traction force, resistance only

Deceleration: $a = -R/m$

Energy: $E_2 = F \times s_{coast}$ (work against resistance)

3. **Braking phase:** Braking force applied

Deceleration: $a = -(F_b + R)/m$

Average Speed & Schedule Speed ■

Average Speed: V_{avg} = Total distance / Total time

Schedule Speed: V_{sch} = Distance / (Running time + Stop time)

■■ **Trap:** Schedule speed < Average speed (includes stop time)

Specific Energy Consumption ■■

$E_{specific}$ = Total energy / (Mass × Distance) = $E/(m \times d)$ Wh/kg-km

Total energy = Acceleration energy + Coasting losses + Braking losses

★ **Regenerative braking:** Returns part of braking energy to supply

5. MEASUREMENTS - BRIDGES & INSTRUMENTS

MEASUREMENT BRIDGES

BRIDGE TYPE	BALANCE CONDITION	MEASURES	FREQUENCY USED
Wheatstone	$R_1/R_2 = R_x/R_3$	Resistance (DC)	DC only
Kelvin	For low R (μΩ)	Low Resistance	DC
Maxwell	R_1 $R_3(1+j\omega C_1R_1)=R_2$ R_x	Inductance L	AC
Anderson	Measures inductance	Inductance L	AC
Schering	$C_x/C_s=R_2/R_1$	Capacitance C	AC (HV)
Hay	Used for high Q	Inductance L (high Q)	AC
Wien	$(1+j\omega C_1R_1)R_2=R_x$	Frequency f	AC

Pointer Instruments - Deflecting Torque ■■

PMMC (Permanent Magnet Moving Coil):

T_d = NBAI (N-m)

Where: N=turns, B=flux density, A=coil area, I=current

Controlling Torque: $T_c = K\theta$ (spring torque)

At equilibrium: NBAI = $K\theta$ ■ $\theta \propto I$ (linear deflection)

Moving Iron Instrument:

$T_d = (dL/d\theta) \times I^2 / 2$ (non-linear)

Deflection $\propto I^2$ (responds to both AC & DC)

Dynamometer (Electrodynamic):

$T_d = kI_1I_2\cos\phi$

Used for power measurement

Sensitivity & Accuracy

Sensitivity: S = Deflection / Applied input (div/V or div/A)

Accuracy: ± % of full scale reading (FSR)

■■ **Trap:** High sensitivity ≠ High accuracy; need good design quality

6. ANALOG & DIGITAL ELECTRONICS

DIODES & RECTIFIERS

Half-Wave Rectifier ■■

Average (DC) Output: $V_{dc} = V_m/\pi = 0.318V_m$

RMS Output: $V_{rms} = V_m/2 = 0.5V_m$

Ripple Factor: $\gamma = \sqrt{(V_{rms}^2 - V_{dc}^2) / V_{dc}^2} \approx 1.21$

Efficiency: $\eta = (V_{dc}^2 / V_{rms}^2) \times 100 \approx 40.6\%$

PIV (Peak Inverse Voltage) = V_m

Full-Wave Rectifier ■■

Average (DC) Output: $V_{dc} = 2V_m/\pi = 0.636V_m$

RMS Output: $V_{rms} = V_m/\sqrt{2} \approx 0.707V_m$

Ripple Factor: $\gamma \approx 0.482$

Efficiency: $\eta \approx 81.2\%$

PIV (Center-tap) = V_m, PIV (Bridge) = $2V_m$

■■ **Trap:** Full-wave bridge has $2V_m$ PIV, not V_m

Bridge Rectifier - Transformer Utilization Factor

TUF = $V_{dc} \times I_{dc}$ / (V_A rated × I_A rated)

For full-wave bridge: TUF ≈ 0.812 (most efficient)

For center-tap: TUF ≈ 0.693

BIPOLAR JUNCTION TRANSISTOR (BJT)

Current Relationships ■■

Emitter Current: $I_E = I_B + I_C$

Current Gain (β): $\beta = I_C / I_B$ (typical 50-200)

Current Gain (α): $\alpha = I_C / I_E = \beta/(\beta+1)$ (typical 0.98-0.99)

Relationship: $\alpha = \beta/(1+\beta)$, $\beta = \alpha/(1-\alpha)$

■■ **Trap:** β >> α; small change in α ■ large change in β

Common Emitter (CE) Configuration ■■

Voltage Gain: $A_v = -g_m \times R_c$ (negative = phase shift)

Current Gain: $A_i = \beta$

Input Impedance: $Z_{in} = \beta \times r_e$ where $r_e = V_T/I_E \approx 26mV/I_E(mA)$

Output Impedance: $Z_{out} \approx R_c$

Common Collector (CC/Emitter Follower) ■

Voltage Gain: $A_v \approx 1$ (unity, no inversion)

Current Gain: $A_i \approx \beta$

High input impedance, Low output impedance

Application: Buffer amplifier, impedance matching

FIELD EFFECT TRANSISTOR (FET)

JFET - Shockley Equation ■■

$I_D = I_{DSS} \times (1 - V_{GS}/V_P)^2$

Where: V_P = pinch-off voltage, V_{GS} = gate-source voltage

Transconductance: $g_m = dI_D/dV_{GS} = -2I_{DSS}(1-V_{GS}/V_P)/V_P$

Voltage Gain: $A_V = -g_m R_D$

■■ **Trap:** g_m varies with V_{GS} (non-linear)

MOSFET - Enhancement Mode ■■

$I_D = (k/2) \times (V_{GS} - V_T)^2$ where $k = \mu \times C_{ox} \times (W/L)$

Threshold Voltage: V_T (typically 1-5V)

No gate current (high input impedance $Z_{in} \approx 10^{12}\Omega$)

Widely used in digital circuits & power electronics

7. POWER ELECTRONICS & DRIVES

POWER SEMICONDUCTOR DEVICES

Silicon Controlled Rectifier (SCR) ■■■

Forward Blocking Voltage: V_{DRO}

Reverse Blocking Voltage: V_{RRM}

Holding Current: I_H (minimum to keep SCR ON)

Latching Current: I_L (trigger current, > I_H)

Forward Current Rating: $I_{F(AV)}$ (max avg forward current)

■■ **Trap:** Latching current > Holding current (opposite of intuition)

★ **Triggering:** Gate pulse width > 100 µs ensures reliable turn-on

Gate Turn-Off Thyristor (GTO) ■

Can be turned off by negative gate pulse (unlike SCR)

Turn-off gain: $\beta_{off} = I_A / I_G$ (1/10 to 1/5)

Application: Choppers, inverters, motor drives

IGBT (Insulated Gate Bipolar Transistor) ■

Combines advantages: MOSFET input + BJT output

Fastest growing device in power electronics

High input impedance, low saturation voltage, fast switching

CONVERTERS: AC-DC, DC-DC, DC-AC

Single-Phase Controlled Rectifier (SCR) ■■

Half-wave (1 SCR):

$V_{avg} = (V_m/\pi) \times (1 + \cos\alpha)$ where α = firing angle

Full-wave (2 SCR, mid-point):

$V_{avg} = (2V_m/\pi) \times \cos\alpha$

Bridge (4 SCR):

$V_{avg} = (2V_m/\pi) \times \cos\alpha$ (same as mid-point)

Firing Angle: $\alpha = 0°$ ■ Full-wave rectifier ($\alpha = 90°$■ DC output = 0)

■■ **Trap:** V_{avg} decreases as α increases (directly proportional to $\cos\alpha$)

DC-DC Chopper (Buck Converter) ■■

Output Voltage: $V_o = V_i \times D$ where D = duty cycle

Duty Cycle: $D = T_{on} / T$ where T_{on} = ON time, T = period

Frequency: $f = 1/T$

Ripple current: $\Delta I = V_i \times D \times (1-D) / (f \times L)$

★ **Control:** Varying D or f changes output voltage

DC-AC Inverter (Square Wave) ■

Output Voltage: $V_o = V_{dc}$ (square wave)

Fundamental Frequency: $f_1 = 1/(2T)$

THD (Square wave): Contains odd harmonics only

$V_1 = (4/\pi) \times V_{dc}$, $V_3 = V_1/3$, $V_5 = V_1/5$, etc.

8. QUANTITATIVE APTITUDE - SHORTCUTS & FORMULAS

TOPIC	FORMULA/METHOD	EXAMPLE	TIP
Percentage	% change = (New - Old)/Old × 100	Price 100→120: 20% increase	Always divide by original
Profit/Loss	Profit% = (SP-CP)/CP × 100	CP=100, SP=120: 20% profit	Loss% formula similar
Simple Interest	SI = (P × R × T) / 100	P=1000, R=5%, T=2 yrs: SI=100	Amount = P+SI
Compound Interest	A = P(1+r/100)<super>n</super>	P=1000, r=10%, n=2: A=1210	CI = A - P
Speed-Time-Distance	Distance = Speed × Time	Speed 60 km/h, time 2h: D=120km	Avg speed = Total D/Total T
Ratio-Proportion	If a:b = c:d, then a×d = b×c	Find x: 2:3 = x:6 ■ x=4	Cross multiply
Average	Avg = Sum of all / Count	2,4,6,8,10: Avg=6	Median useful if outliers
Combinations	C(n,r) = n! / (r!(n-r)!)	C(5,2) = 10	Order doesn't matter
Permutations	P(n,r) = n! / (n-r)!	P(5,2) = 20	Order matters
Probability	P = Favorable / Total	Dice: P(>3) = 3/6 = 0.5	P ranges 0 to 1
Mensuration (Circle)	A = πr<super>2</super>, C = 2πr	r=5: A≈78.5, C≈31.4	Remember π≈22/7
Mensuration (Triangle)	A = ½ × base × height	b=10, h=6: A=30	Heron: s=(a+b+c)/2

TOP 50 FORMULA TRAPS IN APTRANSCO EXAMINATION

■■ **ELECTRICAL FUNDAMENTALS** 1. **RMS vs Average:** For sine: RMS = 0.707 Peak, Average = 0.637 Peak (NOT same!) 2. **Power formulas:** $P = VI$ (general), $P = I^2R$ (DC only), $P = V^2/R$ (R must be constant) 3. **Voltage Divider:** Only works with OPEN-CIRCUIT output (no load current) 4. **Current Divider:** Current goes inversely to resistance (high R ■ low I) 5. **Resistance Series vs Parallel:** Series: $R_{eq} = R_1 + R_2$ | Parallel: $1/R_{eq} = 1/R_1 + 1/R_2$ ■■ **AC CIRCUITS & THREE-PHASE** 6. **Impedance:** $Z = \sqrt{(R^2 + X^2)}$ NOT R + X (must square first) 7. **Line vs Phase voltages (Star):** $V_{line} = \sqrt{3} \times V_{phase}$ (factor √3, not 3) 8. **Line vs Phase current (Delta):** $I_{line} = \sqrt{3} \times I_{phase}$ 9. **Three-phase power:** $P = \sqrt{3}\ VI\cos\phi$ (always √3, not 3) 10. **Leading vs Lagging PF:** Capacitor = leading, Inductor = lagging ■■ **TRANSFORMERS** 11. **Transformation ratio sign:** $VR \approx (I\ R\cos\phi \pm I\ X\sin\phi) / V \times 100$ | (+) lagging, (–) leading 12. **Efficiency vs Regulation:** Both important but DIFFERENT | Efficiency ↓ if losses ↑ | Regulation ↓ if impedance ↓ 13. **Max efficiency condition:** Copper loss = Core loss (happens at fractional load, not full load!) 14. **OC vs SC tests:** OC = core loss, SC = copper loss (not vice versa) 15. **Ideal transformer power:** $P_{in} = P_{out}$ (no losses assumed) ■■ **DC MACHINES** 16. **EMF formula:** $E = K_e \times N \times \Phi$ (proportional to speed & flux) 17. **Back EMF (motor):** $E_b = V_t - I_aR_a$ (opposes applied voltage) 18. **Torque:** $T = K_t \times \Phi \times I_a$ (proportional to flux & current, NOT speed!) 19. **Shunt motor speed:** Increases if field weakened (flux ↓ ■ speed ↑) 20. **Series motor starting:** High current (no load resistance initially), not safe for high-speed applications ■■ **INDUCTION MOTORS** 21. **Synchronous speed:** $N_s = 120f/P$ (not 60f/P for 50 Hz systems!) 22. **Slip:** Use as FRACTION in formulas, not percentage (s = 0.05, not 5%) 23. **Rotor speed:** $N_r = N_s(1 - s)$ (decreases as load increases, s↑) 24. **Rotor frequency:** $f_r = s \times f$ (at no-load: $f_r \approx 0$, at full-load: $f_r \approx$ 2-3 Hz for 50 Hz) 25. **Maximum torque condition:** $s_{max} = R_r/X_r$ (resistance-to-reactance ratio) 26. **Starting vs Full-load torque:** $T_{start} < T_{max} < T_{pull\text{-}out}$ (NOT always maximum at start!) ■■ **POWER SYSTEMS & TRANSMISSION** 27. **Voltage regulation:** Higher impedance ■ Higher regulation (worse performance) 28. **Short vs Long line:** Short (<80 km): ignore shunt capacitance | Long (>250 km): must include distributed parameters 29. **Corona loss:** Exponential increase above critical voltage (NOT linear) 30. **Ferranti effect:** $V_{receiving} > V_{sending}$ at NO-LOAD (due to line capacitance charging) 31. **Surge impedance:** $Z_c = \sqrt{(L/C)}$ (independent of frequency for ideal lines) ■■ **PER-UNIT SYSTEM** 32. **Base values:** Choose S_{base} & V_{base}, then I_{base} & Z_{base} follow (not arbitrary) 33. **Transformer impedance:** Same on both sides in pu (huge advantage over actual values!) 34. **Multiple transformers:** Must convert all to SAME base before adding impedances ■■ **FAULTS & PROTECTION** 35. **Fault types current:** $I_{3\text{-}phase} > I_{LG} \approx I_{LLG} > I_{LL}$ (typical ranking) 36. **Sequence networks:** For LG: All three connected in series | For LL: Only 1 & 2 in series, 0 not used 37. **IDMT relay time:** $t = 0.14/(M^n - 1) \times T_d$ (decreases if fault current ↑) 38. **Plug setting:** $I_{set} = PS \times I_{CTprimary}$ / CT ratio (common mistake: forgetting CT ratio) 39. **Differential protection:** Operates on $I_d = |I_{in} - I_{out}|$ (current difference, not sum) ■■ **CONTROL SYSTEMS** 40. **Gain Margin vs Phase Margin:** Both must be POSITIVE for stability (not just one) 41. **Steady-state error:** Depends on system TYPE and input type (step/ramp/parabolic) 42. **Root locus:** # branches = # poles (not zeros) | # asymptotes = |# poles – # zeros| 43. **System order:** Determined by highest power of s in denominator (NOT related to system TYPE) 44. **Step response overshoot:** Caused by complex poles (NOT real poles) ■■ **MEASUREMENTS & INSTRUMENTS** 45. **Bridge balance:** Condition depends on connection (series/parallel) & frequency (DC/AC) 46. **PMMC deflection:** $\theta \propto I$ (linear) | Moving Iron: $\theta \propto I^2$ (non-linear) 47. **Accuracy vs Sensitivity:** Different concepts! High sensitivity ≠ High accuracy 48. **Instrument error:** Always as % of FULL-SCALE READING, not measured value ■■ **ELECTRONICS** 49. **BJT gains:** $\alpha \approx 0.98\text{-}0.99$, $\beta \approx 50\text{-}200$ ($\alpha << \beta$, but β depends on α: $\beta = \alpha/(1-\alpha)$) 50. **FET transconductance:** Non-linear with V_{GS} (unlike BJT g_m which is more constant)

YOUR 6-DAY FINAL REVISION PLAN (20-26 August 2026)

TODAY (Day 1 - Thursday 21 Aug): Electric Circuits + Control Systems - Morning (2h): DC circuits basics, network theorems, max power transfer - Afternoon (2h): AC fundamentals, RLC, resonance, Q-factor - Evening (2h): Three-phase systems (Star/Delta/Power), 100+ formulas review - Night (1h): Control systems transfer functions, steady-state error, Routh **Day 2 (Friday 22 Aug): Electrical Machines** - Morning (3h): Transformers - EMF, regulation, efficiency, tests - Afternoon (2h): DC machines - EMF, torque, speed, motor/generator equations - Evening (2h): Induction motors - synchronous speed, slip, torque-slip curve - Night (1h): Synchronous machines, V-curves **Day 3 (Saturday 23 Aug): Power Systems** - Morning (2h): Generation factors, load, demand, capacity factors - Mid-morning (2h): Transmission lines (short/medium/long), voltage regulation - Afternoon (2h): Faults (3-phase, LG, LL, LLG), sequence networks - Evening (2h): Protection relays (IDMT, differential, distance), circuit breakers - Night (1h): Stability, swing equation **Day 4 (Sunday 24 Aug): Measurements + Analog/Digital + Power Electronics** - Morning (2h): Measurement bridges (all 7 types), PMMC/Moving Iron/Dynamometer - Mid-morning (1.5h): Oscilloscope, frequency, phase measurement basics - Afternoon (2h): Diodes, rectifiers (half/full wave), ripple factor, efficiency - Mid-afternoon (1.5h): BJT/FET, small-signal parameters, amplifier basics - Evening (1.5h): SCR, GTO, IGBT basics | Choppers, inverters, AC-DC converters **Day 5 (Monday 25 Aug): Common Section (Aptitude + Reasoning + English + GK + IT)** - Morning (1.5h): Aptitude - percentage, SI/CI, speed-distance, ratios, probability - Mid-morning (1.5h): Reasoning - analog, classification, series, coding-decoding - Afternoon (2h): English - grammar rules, vocabulary, reading comprehension (5 passages) - Evening (1.5h): General Awareness + Computer Knowledge - Night (1h): Mock attempt (20-25 random MCQs from all sections) **Day 6 (Final Revision - Tuesday 26 Aug - EXAM DAY MORNING) ONLY 2-3 HOURS BEFORE EXAM:** - Read this one-page ultra-revision sheet (see next page) - Quick scan of priority formulas (■■■ only) - Top 20 formula traps - No new topics, NO lengthy studying - Eat well, stay hydrated, light mental exercise - Arrive 30 min early to exam center **EXAM STRATEGY (2 hours, 100 questions):** 1. **First 5 min:** Read question paper, identify easy questions 2. **Next 90 min:** Attempt questions in order of difficulty (easy first) - Electrical core (70 marks): Target 60+ marks ✓ (90 min) - Common section (30 marks): Target 24+ marks ✓ (30 min) 3. **Final 5 min:** Review marked answers 4. **No negative marking:** Attempt ALL questions (even guesses) ✓ **Priority Focus (These 20% of formulas will likely give 80% of marks):** - 3-phase power calculation - Transformer voltage regulation & efficiency - Induction motor slip & torque - Synchronous speed formula - Transmission line regulation - Fault current (3-phase & LG) - Control system steady-state error - Rectifier output DC & ripple factor - IDMT relay operating time - Power factor correction kVAR

ONE-PAGE ULTRA-REVISION SHEET (Keep This With You!)

■■ **ULTRA-QUICK FORMULAS (Most Tested) 3-PHASE POWER:** P = √3 VIcosϕ (W) | Q = √3 VIsinϕ (VAR) | S = √3 VI (VA) **TRANSFORMER:** VR ≈ (IR cosϕ + IX sinϕ)/V × 100 | Sign: (+) lag, (–) lead **DC MOTOR:** $E_b = V_t - I_a R_a$ | T = K × Φ × I_a | N = $(E_b)/(K \times \Phi)$ **INDUCTION MOTOR:** N_s = 120f/P | s = $(N_s - N_r)/N_s$ | f_r = sf **MAX TORQUE (IM):** T_{max} occurs at s = R_r/X_r **TRANSMISSION LINE:** VR = I(R cosϕ + X sinϕ)/V (short line) | Efficiency η = $P/(P + I^2R)$ **FAULT CURRENT:** 3-phase: $I_f = V/Z_1$ | LG: $I_f = 3V/(Z_0+Z_1+Z_2)$ **PER-UNIT:** $I_{base} = S_{base}/(\sqrt{3} \times V_{base})$ | $Z_{base} = V_{base}^2/S_{base}$ **CONTROL:** e_{ss}(step) = $1/(1+K_p)$ | e_{ss}(ramp) = $1/K_v$ | K_p=G(0), K_v=lim[sG] **RECTIFIER:** Half-wave: $V_{avg} = V_m/\pi$ | Full-wave: $V_{avg} = 2V_m/\pi$ | γ = 1.21 (HW), 0.482 (FW) **BJT:** β = I_c/I_B | α = β/(1+β) | $I_E = I_B + I_C$ **FET SHOCKLEY:** $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ ■■ **CRITICAL NUMERICAL VALUES (Memorize!)** √2 ≈ 1.414 | √3 ≈ 1.732 | 1/√3 ≈ 0.577 | π ≈ 3.14 RMS (sine) = 0.707 × Peak | Average (sine) = 0.637 × Peak | Form Factor = 1.11 At 50 Hz: 2-pole=3000rpm | 4-pole=1500rpm | 6-pole=1000rpm | 8-pole=750rpm sin60°=0.866, cos60°=0.5 | sin30°=0.5, cos30°=0.866 | tan45°=1, tan60°=1.73 ■■ **TOP 10 FORMULA TRAPS TO AVOID** 1. RMS ≠ Average (off by 1.11×) 2. VR sign: + for lagging, – for leading 3. Slip as FRACTION not percentage (0.05, not 5%) 4. Line voltage Star = √3 × Phase (factor √3, not 3) 5. 3-phase power ALWAYS √3 × V × I × cosϕ 6. Torque ∝ Φ × I_a, NOT speed 7. Max efficiency @ Copper loss = Core loss 8. Fault current: LG > LLG > 3-phase > LL (typical) 9. IDMT time decreases as M (fault ratio) increases 10. β ≠ α (β >> α, related by β = α/(1–α)) ■■ **LAST-MINUTE EXAM TIPS (2 HOURS ONLY) ✓ Electrical (70 marks - 90 min):** • Q1-10 (Circuits): Max Power Transfer, Thevenin, 3-phase basics • Q11-25 (Machines): Transformer VR, IM slip, DC motor speed, Sync impedance • Q26-35 (Power Systems): Transmission VR, fault current, protection relay time • Q36-45 (Control): Steady-state error, Routh stability, Bode margins • Q46-55 (Util/Measurements): Traction speed, bridge balance, IDMT • Q56-65 (Electronics): Rectifier ripple, BJT gains, FET Shockley • Q66-70 (Power Elec): SCR characteristics, chopper duty cycle, inverter freq ✓ **Common (30 marks - 30 min):** • Q71-75 (Reasoning): Analogies, classification, series → 2 min/Q • Q76-82 (Aptitude): Percentage, ratio, SI/CI, speed-dist → 2 min/Q • Q83-87 (English): Grammar, vocab, comprehension → 2 min/Q • Q88-92 (GK): Current affairs, basic facts → 1 min/Q • Q93-100 (IT): OS, spreadsheet basics, networking → 1 min/Q ✓ **If stuck on a question:** • Use dimensional analysis (check units match) • Estimate answer (eliminate obvious wrong options) • Guess if <30 sec already spent (no negative marking) • Flag & return only if time permits ✓ **Final 5 minutes:** • Verify answered all 100 questions • Quick mental check: Are my answers reasonable? • Don't change answers unless absolutely certain • Deep breath, hand in paper confidently ■■ **YOU HAVE STUDIED HARD. BELIEVE IN YOUR PREPARATION. ✓ Execute the plan, avoid the traps, attempt all questions. TARGET: 60+ electrical + 24+ common = 84+ total marks ✓** ■■■

FINAL EXAMINATION STRATEGY & SUCCESS CHECKLIST

EXAMINATION DAY CHECKLIST (26 AUGUST 2026) 1-2 HOURS BEFORE EXAM: ■ Eat light, balanced meal (carbs + protein, no heavy food) ■ Drink water (stay hydrated, brain works better) ■ Review ultra-revision sheet (this page only, NO new content) ■ Scan top 20 formula traps ■ Light walking/stretching (keep blood circulation) ■ Relax mentally - your preparation is complete ■ Arrive at exam center 30 minutes early ■ Check: Admit card, ID, pens, rough paper **IN THE EXAM HALL (FIRST 5 MINUTES):** ■ Read all 100 questions quickly (30 seconds per question) ■ Identify EASY questions (mark mentally) ■ Identify HARD questions (mark for later) ■ Categorize: Very easy (3 min total) | Easy (5 min) | Medium (8 min) | Hard (skip if time) ■ DO NOT start with hard questions! ■ Strategy: Attempt 80% of questions with 95% accuracy > 100% questions with 60% accuracy **TIME ALLOCATION (2 Hours = 120 Minutes):** Section | Marks | Time | Strategy ■■ Electric Circuits | 10 | 15 min | Basics: Ohm, Power, Voltage divider → EASY Networks | 10 | 20 min | Theorems, max power transfer, 3-phase Machines | 15 | 25 min | Transformer VR, IM slip, DC motor → HIGHEST VALUE Power Systems | 18 | 30 min | Transmission VR, faults, relays → MOST FORMULAS Control | 12 | 15 min | Error constants, IDMT, Routh Others | 5 | 10 min | Measurements, basic concepts ■■ Common Section | 30 | 30 min | Aptitude (8 min) | Reasoning (8 min) | English (8 min) | Others (6 min) ■■ TOTAL | 100 | 120 min | ✓ Buffer time included **QUESTION ANSWERING TECHNIQUE:** For NUMERICAL questions: 1. Write given data clearly 2. Identify formula needed 3. Substitute values carefully 4. Calculate step-by-step 5. Check dimensional analysis (units must match) 6. Round appropriately (usually 2-3 significant digits) 7. Check if answer is reasonable (use common sense) For CONCEPTUAL MCQs: 1. Read question carefully (don't miss "NOT", "EXCEPT", etc.) 2. Eliminate obviously wrong options (save 50% of options) 3. Consider trap options (common mistakes) 4. Think of real-world example if applicable 5. Choose most precise answer (not close, but exact) For APTITUDE questions: 1. Convert word problem to equation 2. Use shortcut methods (not long calculations) 3. Estimate answer first (check if your calc matches estimate) 4. If calculation is taking >2 min, likely wrong approach **HANDLING DIFFICULT QUESTIONS:** If stuck for >1 minute: 1. Skip it (mark clearly for return) 2. Come back only if time permits 3. Make educated guess (better than blank, no negative marking) 4. Don't leave ANY question unanswered If calculation is complex: 1. Check if formula can be simplified 2. Use approximations ($\pi \approx 22/7$, $\sqrt{2} \approx 1.4$) 3. Try dimensional analysis (check units) 4. If still stuck, guess and move on **MENTAL BLOCKS DURING EXAM:** If you forget a formula: • Don't panic - derive it if possible • Remember what TYPE of formula (proportional, inverse, exponential) • Check answer using dimensional analysis • Make your best educated guess If nervous/anxious: • Take 3 deep breaths • Remember: You have prepared well • Focus on next question only, not overall • This is just one exam, not your whole career If running out of time: • Switch to guessing mode with logic • Eliminate worst 2 options, choose between remaining 2 • Answer EASY questions in remaining time • No blank answers (even guesses help with no negative marking) **FINAL 10 MINUTES STRATEGY:** With 10 minutes left: 1. If you have attempted 80+ questions: GOOD, verify answers 2. If you have 10-15 unanswered: Make EDUCATED GUESSES (think logically) 3. If time up: Submit whatever you have (incomplete answer better than blank) With 5 minutes left: 1. Quick scan: Any obvious errors? Fix if <30 sec 2. Any blanks remaining? Fill with best guess 3. All 100 answered? YES ✓ → Submit confidently With 1 minute left: 1. STOP calculating 2. Submit paper 3. Don't second-guess yourself 4. DONE! ✓ **AFTER EXAM:** • Don't discuss with others (ruins confidence for other exams) • Rest well (recovery > rehashing) • Wait for results (no use worrying) • Learn from experience for next opportunity ■■ **SUCCESS MANTRA FOR APTRANSCO AEE ELECTRICAL 2026:** "I have studied all formulas, understood all concepts, and prepared for all traps. I will attempt all questions confidently. I will not leave any blank. My target is 84+ marks, and I will achieve it through focused strategy and no panic. LET'S GO! ✓" ■■■